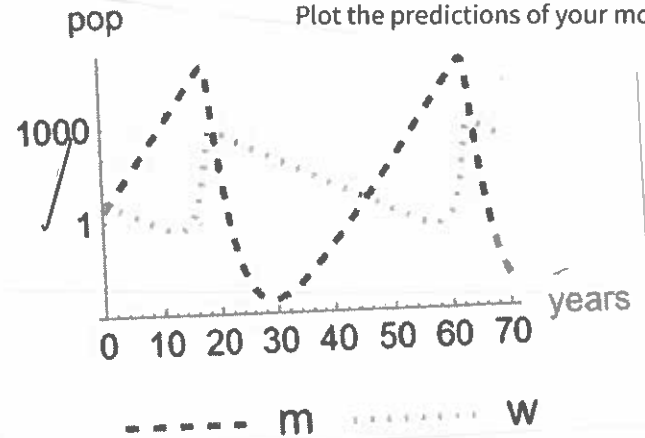


Wiki: 4 calves every 3 years "moose"
 : life expectancy 20 years (moose) $\alpha = (\frac{1}{2} \cdot \frac{4}{3} - \frac{1}{20}) = \frac{2}{3} - \frac{1}{20}$
 : wolf life expectancy 5 years $\delta = \frac{1}{5}$
 If our island has an eq. population of 5000 moose
 and 100 wolves then $w_{eq} = 100 = \alpha/\beta$ and $m_{eq} = 5000 = \delta/\beta$
 $\beta = \frac{\alpha}{100}$ $\delta = \frac{1}{5}$

Name

- Estimate parameters in the Lotka-Volterra model $m' = \alpha m - \beta m w$ for moose (m) and wolves (w).
 $w' = -\gamma w + \delta m w$

Plot the predictions of your model.



1. Show some reasoning to estimate birth and death rates for rabbits then write down and solve a simple ODE modeling an initially small population of rabbits $r(t)$ on a lush fertile island. Discuss the results.

10 years life exp.
 6 kits per litter
 4 litters per year

Assume $\frac{1}{2}$ of rabbits are female
 time in years

$$r'(t) = (\frac{1}{2}(6)(4) - \frac{1}{10})r(t) = 11.9 r(t)$$

$$r(t) = r(0)e^{11.9t}$$

$$r(1) = 147,267 r(0)$$

Rabbits can multiply their population by more than 100,000 in a year!!